

Assignment 1 – Initiate

Applied Deep Learning

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1. Project Title

Prediction of Lift (C_L) and Drag (C_D) Coefficients for Airfoils using Deep Learning

2. Overview and Goals

The objective of this project is to develop a deep learning-based surrogate model capable of predicting the aerodynamic coefficients — specifically the lift coefficient (C_L) and drag coefficient (C_D) — for various airfoil geometries and flow conditions.

Traditional Computational Fluid Dynamics (CFD) tools such as XFOIL or OpenFOAM can accurately compute aerodynamic coefficients, but they are computationally expensive and time-consuming when used for large parameter studies or optimization loops. This project aims to explore whether a neural network can approximate these aerodynamic behaviors with significantly reduced computational cost.

By training a neural network on precomputed aerodynamic data, we can quickly estimate C_L and C_D for new airfoil shapes or flow conditions, thereby accelerating design-space exploration and optimization workflows.

3. Project Type

Bring Your Own Method

This project uses an existing publicly available dataset (DeepLearWing) and applies a Multilayer Perceptron (MLP) model to predict aerodynamic coefficients. The focus is on implementing, training, and evaluating a neural network model rather than collecting new data.

4. Dataset Description

Dataset: [DeepLearWing – Airfoil Aerodynamics Dataset \(Kaggle\)](#)

Overview: The DeepLearWing dataset contains thousands of airfoil samples generated via CFD simulations. Each sample describes the airfoil geometry, flow conditions, and corresponding aerodynamic coefficients.

Category	Example Features	Description
Geometry	Coordinates (x, y), thickness, camber,	Defines airfoil shape
Flow Conditions	Angle of Attack (AoA), Reynolds no., Mach no.	Describes flight condition
Outputs	Lift (C_L), Drag (C_D)	Target variables for regression

Motivation: This dataset provides both geometric and physical inputs, which makes it suitable for supervised regression tasks. It enables a compact MLP to learn aerodynamic trends without requiring CFD simulations during inference.

5. Proposed Methodology

Data Preprocessing

- Normalize and scale geometric and flow condition inputs.
- Split dataset into training (70%), validation (15%), and test (15%) sets.

Model Architecture

- **Model Type:** Multilayer Perceptron (fully connected network)
- **Hidden Layers:** 3–5 layers, 128–512 neurons each
- **Activation:** ReLU
- **Optimizer:** Adam
- **Loss Function:** Mean Squared Error (MSE)
- **Metrics:** Mean Absolute Error (MAE), Coefficient of Determination (R^2)

Training and Evaluation

- Perform hyperparameter tuning (learning rate, batch size, dropout).
- Train until convergence with early stopping to prevent overfitting.
- Evaluate model on unseen airfoils to validate generalization.

Expected Outcome

The trained neural network should predict C_L and C_D values close to reference CFD data, with errors typically within 5–10%. The resulting model will act as a fast, approximate surrogate for aerodynamic evaluation.

6. Work Breakdown Structure and Time Plan

Phase	Task	Estimated Time
Literature Review	Study prior works and related datasets	6–8 hrs
Dataset Exploration	Inspect, preprocess, and normalize DeepLearWing data	8–10 hrs
Model Design	Build baseline MLP model in PyTorch or TensorFlow	10–12 hrs
Training and Hyperparameter Tuning	Experiment with learning rate, batch size, layers	15–20 hrs
Model Evaluation	Compute metrics, visualize results, compare with literature	8–10 hrs
Application Demo	Create Jupyter Notebook or simple web demo for predictions	6–8 hrs
Report and Presentation	Write final report and prepare slides	8–10 hrs

Total Estimated Effort: 60–70 hours

7. References

- Moin, H., Khan, H. Z. I., Mobeen, S., & Riaz, J. (2021). *Airfoil’s Aerodynamic Coefficients Prediction using Artificial Neural Network*. arXiv preprint arXiv:2109.12149. Available at: <https://arxiv.org/abs/2109.12149>
- Wang, X., Qian, W., Zhao, T., He, L., Chen, H., Sun, H., Tian, Y., & Cui, J. (2024). *Fast Prediction of Airfoil Aerodynamic Characteristics Based on a Combined Autoencoder*. *Symmetry*, 16(7), 791. Available at: <https://www.mdpi.com/2073-8994/16/7/791>